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SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, COATED, GLASS, SILICONE RUBBER-COATED

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for silicone rubber-coated glass cloth to be used in the fabrication of stovepipe shields.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-P-1136	-	Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-Y-1140	-	Yarn, Cord, Sleeving, Cloth, and Tape - Glass
MIL-C-20079	-	Cloth, Glass; Tape, Textile Glass; and Thread, Glass

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-406 - Plastics; Methods of Testing
FED-STD-601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specificiations and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard shade sample. The coated cloth shall match the standard sample for shade only (see 6.3).

3.2 Materials (see 6.4).

* 3.2.1 Base cloth. The base cloth shall be glass cloth conforming to class C, Form 4, Cloth No. 128 of MIL-Y-1140 or type I, class 2 of MIL-C-20079 except the requirement for resin treatment shall not apply.

3.2.2 Coating compound. The coating compound shall be a composition of a silicone rubber suitably compounded and pigmented to meet the requirements specified herein.

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3.3 Coated cloth. The base cloth shall be coated on both sides with approximately equal amounts of the coating compound specified in 3.2.2. The coated cloth shall be vulcanized and shall conform to the requirements of table I when tested as specified in 4.2.2.2.

* TABLE I. Physical requirements of coated cloth

Characteristic	Requirement	
	Minimum	Maximum
Weight, oz/sq. yd. (g/m^2)	18.0 (610)	21.0 (712)
Breaking strength, pounds (newton):		
Warp	140 (623)	
Filling	120 (534)	
Tearing strength, grams (millinewton):		
Warp	6000 (58842)	
Filling	6000 (58842)	
Hydrostatic resistance, p.s.i. (kPa):		
Initial	180 (1241)	
After accelerated weathering	170 (1172)	
Stiffness, warp direction, centimeters:		
At 75°F (24°C)		21.0
At minus 60°F (16°C)		22.0
After heat stability 1/		23.0
Flame resistance, after flame, seconds-		
Initial:		
Warp		13
Filling		13
After accelerated weathering:		
Warp		13
Filling		13
Abrasion resistance	Pass 2/	
Blocking, scale rating		3

1/ There shall be no appreciable change in color in any of the specimens when tested as specified in 4.2.2.2. Appreciable change in color means a change which is immediately noticeable in comparing the test specimen

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with the original comparison specimen. If closer inspection or a change of angle of light is required to make a slight change in color, the change is not considered appreciable.

2/ The coating shall not be worn through to the base cloth on any of the specimens when tested as specified in 4.2.2.2.

3.4 Width. The minimum width of the coated cloth shall be 36 inches (914 mm).

3.5 Length and put-up. The coated cloth shall be put up as specified in PPP-P-1136 in rolls not more than 100 yards (91 m) nor less than 75 yards (69 m) in length. No roll shall contain more than 3 pieces, none of which is less than 15 yards (14 m) in length. The ends of the pieces shall be overlapped, not joined by a seam.

3.6 Color. The color of the coated cloth shall be Olive Drab 209 and shall match the standard shade sample (see 6.3).

* 3.6.1 Matching. The color shall match the standard sample under artificial daylight having a color temperature of 7000 ± 500 kelvin and shall be a good approximation of the standard sample under incandescent lamplight at 2850 ± 100 kelvin.

3.7 Workmanship. The finished coated cloth shall conform to the quality of product established by this specification. The occurence of defects shall not exceed the specified acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2 End item inspection.

4.2.2.1 End item examination. The end item shall be examined in accordance with the classification of defects at the inspection levels and acceptable quality levels (AQLs) set forth below. The lot size for purposes of determining the sample size shall be expressed in units of yards for the examinations in 4.2.2.1.1 and 4.2.2.1.2.

4.2.2.1.1 Yard-by-yard visual examination. The required yardage of each roll shall be examined on one side only (however, the sample shall be alternated on every other roll examined) and visual defects classified as listed below. The same yardage shall be given a through-light inspection for pinholes, and lightly coated areas. The through-light shall be performed in accordance with MIL-STD-1487. The defects listed below shall be counted, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warp-wise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The acceptable quality level for this examination, including the pinholes and lightly coated areas found in the through-light examination, shall be 2.5 defects per 100 units (yards). The inspection level shall be level II of MIL-STD-105. The number of rolls from which the sample is to be selected shall be in accordance with table II. An approximately equal number of yards shall be examined in each roll sampled.

Examine	Defect
Coating	Any uncoated area. Any thinly coated area evidenced by light area or windows. Any lump or heavily coated area. Any blister, tunnel or delamination of coating. Any pinhole. Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure.
Color	Color not as specified, off shade, uneven or mottled.
Width	Width less than minimum specified.
Workmanship	Any cut, hole or tear. Any stains or streaks other than coating or pigment clearly visible 1/ and exceeding one half inch in combined length and width. Curled, folded, scalloped or rolled edge. Not clean. Objectionable odor. 2/ Any abrasion exposing base cloth.

1/ At normal inspection distance (approximately 3 feet).

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2/ Aromatic odors usually associated with specific types of coating compounds or solvents shall not be regarded as objectionable.

TABLE II

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1,200 inclusive 1/	3	0
1,201 up to and including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.2.2.1.2 Dimensional examination for length.

4.2.2.1.2.1 Examination for length of individual roll. Each individual roll in the sample shall be examined for the defects listed below. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table II.

Any roll containing more than 3 pieces.
Any piece in roll less than 15 yards.
Any roll with a gross length of less than 75 yards or more than 100 yards.
Any roll with a gross length more than 2 yards below that marked on ticket.

4.2.2.1.2.2 Examination for total yardage in sample. The rolls examined shall be those selected for examination of individual rolls as specified in 4.2.2.1.2.1. The lot shall be unacceptable if the total of the gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the tickets.

4.2.2.2 End item testing. The methods of testing specified in FED-STD-191, whenever applicable and as listed in table IV, shall be followed. The requirements for all tests are referenced in table I. The physical values specified apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods, except for abrasion, blocking and flame resistance, for which these values apply to each single determination. The sample unit shall be 3 continuous yards, full width, of the coated cloth. The lot size shall be expressed in units of 1 linear yard and the sample size (number of sample units) shall be as specified in table III. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The number of determinations per sample unit, as well as the reporting of test results, shall be as required by the referenced test method except as otherwise indicated in table IV.

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TABLE III

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE IV. Instructions for testing

Characteristic	Test method	No. of determinations per sample unit	Results reported to the nearest
Weight	5041	5	0.1 ounce
Breaking strength	5102	5 (each direction)	1.0 pound
Tearing strength	ASTM D-1424	5 (each direction)	100 grams
Hydrostatic resistance:			
Initial	5512	5	2 p.s.i.
After accelerated weathering	5804 1/ 5512	5	2 p.s.i.
Stiffness: (warp direction only)			
At 75°F.	5204	5	0.1 centimeter
At minus 60°F.	5204 2/	5	0.1 centimeter
After heat stability	4.3.1 and 5204	5	0.1 centimeter
Flame resistance: 3/			
Initial	5903	5 (each direction)	0.2 second
After accelerated weathering	5804 1/ 5903 3/	5 (each direction)	0.2 second
Abrasion resistance	4.3.2	3	Pass or fail
Blocking	5872	1	Scale rating

1/ The exposure period shall be 100 hours.

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2/ The specimens and the instrument shall be conditioned for 4 hours at minus 60° ± 2°F before testing at that temperature.

3/ Except that the glow time and char length need not be reported.

* 4.2.3 Packaging inspection. Examination shall be made in accordance with the provisions of PPP-P-1136 to determine that the packing and marking comply with the section 5 requirements.

4.3 Methods of inspection. Results of physical tests obtained under testing conditions defined in FED-STD-191, FED-STD-406, or FED-STD-601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

4.3.1 Heat stability. Five specimens 1-1/4 by 8 to 12 inches with the long dimensions parallel to the warp shall be placed flat on a clean, asbestos board in a muffle furnace or in a suitably controlled high temperature oven for a period of 4 hours at a temperature of between 575°F and 600°F. After exposure, the specimens shall be removed and brought to standard conditions before testing. The stiffness shall be determined according to Method 5204 of FED-STD-191. The exposed specimens shall be visually compared with an equivalent area of unexposed material for colorfastness. Color change shall be reported as pass or fail.

4.3.2 Abrasion resistance. The abrasion resistance shall be determined according to Method 5302 of FED-STD-191 except that 3 specimens shall be tested. The three specimens shall be selected across the width of the sample unit so that two of these specimens are obtained from opposite sides and one from the middle. The diaphragm shall be solid rubber (no contact pin) 0.050 to 0.058 inch thick. The abradant shall be "fine" emery cloth and the abradant arm shall be weighted with a 1/2-pound weight. The air pressure shall be 1.2 p.s.i. and the abrasion shall be multidirectional. The specimen shall be abraded on either side for 20 cycles and then visually examined for exposure of the base cloth.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

* 5.1.1 Levels A and C. The coated cloth shall be put up on rolls (see 3.5) together with a full width interleaved sheet of polyethylene film running the entire roll length, and packaged in accordance with the applicable requirements for class 6 fabrics of PPP-P-1136. The polyethylene flexible film shall be 0.003 inch thick conforming to type I, class 1, grade A, B or C, finish 1 of L-P-378.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth is intended to be used in the manufacture of stovepipe shields for tentage.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
* (b) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard shade sample. A standard shade sample of the coated cloth may be obtained from the procuring activity issuing the invitation for bids.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

* 6.5 Metric equivalents. Metric equivalents, indicated in parentheses in section 3 of this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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